Topic: Changing the temperature boundaries

Question: Find a solution to the partial differential equation.

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \text{ for } 0 \leq x \leq 10$$

$$u(0,t) = 5,\ u(10,t) = 15$$

$$u(x,0) = x + 10$$

Answer choices:

A $$10 + x + \sum_{n=1}^{\infty} (1 - (-1)^n)\frac{10}{n\pi} \sin\left(\frac{n\pi x}{10}\right) e^{-\left(\frac{n\pi}{10}\right)^2 t}$$

B $$10 + x + \sum_{n=1}^{\infty} ((-1)^n - 1)\frac{10}{n\pi} \cos\left(\frac{n\pi x}{10}\right) e^{-\left(\frac{n\pi}{10}\right)^2 t}$$

C $$5 + x + \sum_{n=1}^{\infty} (1 - (-1)^n)\frac{10}{n\pi} \sin\left(\frac{n\pi x}{10}\right) e^{-\left(\frac{n\pi}{10}\right)^2 t}$$

D $$5 + x + \sum_{n=1}^{\infty} ((-1)^n - 1)\frac{10}{n\pi} \cos\left(\frac{n\pi x}{10}\right) e^{-\left(\frac{n\pi}{10}\right)^2 t}$$

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Solution: C

First we'll find the function modeling equilibrium temperature.

$$u_E(x) = T_1 + \frac{T_2 - T_1}{L}x$$

$$u_E(x) = 5 + \frac{15 - 5}{10}x$$

$$u_E(x) = 5 + x$$

Next, we'll find the coefficient B_n.

$$B_n = \frac{2}{L}\int_0^L (f(x) - u_E(x))\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{10}\int_0^{10} (x + 10 - 5 - x)\sin\left(\frac{n\pi x}{10}\right)\,dx$$

$$B_n = \frac{1}{5}\int_0^{10} 5\sin\left(\frac{n\pi x}{10}\right)\,dx$$

$$B_n = -\frac{10}{n\pi}\cos\left(\frac{n\pi x}{10}\right)\Big|_0^{10}$$

$$B_n = -\frac{10}{n\pi}(\cos(n\pi) - 1)$$

$$B_n = -\frac{10}{n\pi}((-1)^n - 1)$$

$$B_n = \frac{10}{n\pi}(1 - (-1)^n)$$

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Then the solution to the equation is

$$u(x,t) = u_E(x) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

$$u(x,t) = 5 + x + \sum_{n=1}^{\infty} (1-(-1)^n)\frac{10}{n\pi} \sin\left(\frac{n\pi x}{10}\right) e^{-\left(\frac{n\pi}{10}\right)^2 t}$$

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Question: Find a solution to the partial differential equation.

$$\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2} \text{ for } 0 \leq x \leq 2\pi$$

$$u(0,t) = 20,\ u(2\pi, t) = 40$$

$$u(x,0) = 20$$

Answer choices:

A $$u(x,t) = 20 + \frac{10}{\pi}x + \frac{40}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\sin\left(\frac{nx}{2}\right)e^{-n^2t}$$

B $$u(x,t) = 20 + \frac{10}{\pi}x + \frac{20}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\sin\left(\frac{nx}{2}\right)e^{-n^2t}$$

C $$u(x,t) = 20 + \frac{10}{\pi} + \frac{40}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\left(\frac{nx}{2}\right)e^{-4n^2t}$$

D $$u(x,t) = 20 + \frac{10}{\pi} + \frac{20}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\sin\left(\frac{nx}{2}\right)e^{-4n^2t}$$

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Solution: A

First we'll find the function modeling equilibrium temperature.

$$u_E(x) = T_1 + \frac{T_2 - T_1}{L}x$$

$$u_E(x) = 20 + \frac{40 - 20}{2\pi}x$$

$$u_E(x) = 20 + \frac{10}{\pi}x$$

Next, we'll find the coefficient B_n.

$$B_n = \frac{2}{L}\int_0^L (f(x) - u_E(x))\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{1}{\pi}\int_0^{2\pi}\left(20 - 20 - \frac{10}{\pi}x\right)\sin\left(\frac{nx}{2}\right)\,dx$$

$$B_n = -\frac{10}{\pi^2}\int_0^{2\pi} x\sin\left(\frac{nx}{2}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(nx/2)\,dx$, and $v = -(2/n)\cos(nx/2)$.

$$B_n = -\frac{10}{\pi^2}\left(-\frac{2}{n}x\cos\left(\frac{nx}{2}\right)\Big|_0^{2\pi} + \frac{2}{n}\int_0^{2\pi}\cos\left(\frac{nx}{2}\right)\,dx\right)$$

$$B_n = \frac{20}{n\pi^2}x\cos\left(\frac{nx}{2}\right) - \frac{40}{(n\pi)^2}\sin\left(\frac{nx}{2}\right)\Big|_0^{2\pi}$$

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$$B_n = \frac{20}{n\pi^2}(2\pi)\cos\left(\frac{2n\pi}{2}\right) - \frac{40}{(n\pi)^2}\sin\left(\frac{2n\pi}{2}\right)$$

$$-\left[\frac{20}{n\pi^2}(0)\cos\left(\frac{n(0)}{2}\right) - \frac{40}{(n\pi)^2}\sin\left(\frac{n(0)}{2}\right)\right]$$

$$B_n = \frac{40}{n\pi}\cos(n\pi) - \frac{40}{(n\pi)^2}\sin(n\pi)$$

$$B_n = \frac{40}{n\pi}(-1)^n$$

Then the solution to this equation is

$$u(x,t) = u_E(x) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

$$u(x,t) = 20 + \frac{10}{\pi}x + \frac{40}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\sin\left(\frac{nx}{2}\right)e^{-n^2 t}$$

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Question: Find a solution to the partial differential equation.

$$\frac{\partial u}{\partial t} = 3\frac{\partial^2 u}{\partial x^2} \text{ for } 0 \leq x \leq 5$$

$$u(0,t) = 10,\ u(5,t) = 20$$

$$u(x,0) = 5 + x$$

Answer choices:

A $$u(x,t) = 10 + 2x + \frac{10}{\pi}\sum_{n=1}^{\infty}\frac{1+(-1)^{n+1}}{n}\sin\left(\frac{n\pi x}{5}\right)e^{-\left(\frac{n\pi}{5}\right)^2 t}$$

B $$u(x,t) = 2x + \frac{10}{\pi}\sum_{n=1}^{\infty}\frac{1+(-1)^{n+1}}{n}\sin\left(\frac{\pi n x}{5}\right)e^{-3\left(\frac{n\pi}{5}\right)^2 t}$$

C $$u(x,t) = 10 + 2x + \frac{10}{\pi}\sum_{n=1}^{\infty}\frac{2(-1)^{n}-1}{n}\sin\left(\frac{n\pi x}{5}\right)e^{-\left(\frac{n\pi}{5}\right)^2 t}$$

D $$u(x,t) = 10 + 2x + \frac{10}{\pi}\sum_{n=1}^{\infty}\frac{2(-1)^{n}-1}{n}\sin\left(\frac{n\pi x}{5}\right)e^{-3\left(\frac{n\pi}{5}\right)^2 t}$$

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Solution: D

First we'll find the function modeling equilibrium temperature.

$$u_E(x) = T_1 + \frac{T_2 - T_1}{L}x$$

$$u_E(x) = 10 + \frac{20 - 10}{5}x$$

$$u_E(x) = 10 + 2x$$

Next, we'll find the coefficient B_n.

$$B_n = \frac{2}{L}\int_0^L (f(x) - u_E(x))\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{5}\int_0^5 (5 + x - 10 - 2x)\sin\left(\frac{n\pi x}{5}\right)\,dx$$

$$B_n = -\frac{2}{5}\int_0^5 (x + 5)\sin\left(\frac{n\pi x}{5}\right)\,dx$$

Use integration by parts with $u = x + 5$, $du = dx$, $dv = \sin(n\pi x/5)\,dx$, and $v = -(5/n\pi)\cos(n\pi x/5)$.

$$B_n = -\frac{2}{n\pi}\left(-(x + 5)\cos\left(\frac{n\pi x}{5}\right)\Big|_0^5 + \int_0^5 \cos\left(\frac{n\pi x}{5}\right)\,dx\right)$$

$$B_n = \frac{2(x + 5)}{n\pi}\cos\left(\frac{n\pi x}{5}\right) - \frac{10}{(n\pi)^2}\sin\left(\frac{n\pi x}{5}\right)\Big|_0^5$$

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$$B_n = \frac{20}{n\pi}\cos(n\pi) - \frac{10}{(n\pi)^2}\sin(n\pi) - \frac{10}{n\pi}$$

$$B_n = \frac{20}{n\pi}(-1)^n - \frac{10}{n\pi}$$

$$B_n = \frac{10}{n\pi}(2(-1)^n - 1)$$

Then the solution to this equation is

$$u(x,t) = u_E(x) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

$$u(x,t) = 10 + 2x + \frac{10}{\pi}\sum_{n=1}^{\infty} \frac{2(-1)^n - 1}{n} \sin\left(\frac{n\pi x}{5}\right) e^{-3\left(\frac{n\pi}{5}\right)^2 t}$$

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